

corporate bonds than for equities and also the bid-ask spread. Investors seeking to purchase corporate bonds will often have to find a dealer with an existing inventory and then additionally pay the ask price. Investors seeking to sell corporate bonds will need to find a dealer willing to take on inventory and then sell the bonds at the bid price. The corresponding risk-premium of a corporate bond can be dramatically reduced by these transaction costs if the investor does not hold the bond for its remaining duration. We do expect corporate bond transaction costs to decrease with increased market transparency. We did not find a large relationship between corporate bonds and order size as has been found with equities. Transaction costs in the corporate bond market appear to be related to the competitiveness of the market and number of corresponding dealers.

Commodities. Commodity transaction costs were on average about –50% lower than large cap stocks. Much of this cost was due to the bid-ask spread rather than due to the transaction size. We also found that trading costs across different commodities varied greatly. For example, precious metals had costs that were much different than fossil fuels such as oil and natural gas, and were much different than agricultural goods such as corn, sugar, wheat, etc. Commodity prices did not appear to be strongly related to actual transaction size as it is for equities. Thus, a structural difference between equities and commodities.

Currency. Currency trading (e.g., exchange rates, FX) was –63% less than large cap stocks. The largest component of the FX trading cost was the market spread. We did not find as large a relationship between trading cost and transaction value for currencies as we observed for stocks. But investors did transact at the full spread rather than transacting within the spread as is often accomplished in the equities markets. The market structure for FX trading is much different than the equities markets.

These results are shown in [Table 12.2b](#).

A major finding between the cost structure with equities and other asset classes is that price sensitivity to the underlying order size (e.g., price elasticity) is much more instrument specific than it is for stocks. We found the parameters a_1 , a_2 , a_3 , a_4 and b_1 were relatively stable across stocks in our equity market grouping (e.g., US-LC, US-SC, Europe, etc.), but these parameters did vary by instruments in other asset classes. For example, the model parameters could be much different for a liquid broad market ETF compared to a specific factor ETF such as a dividend yielding ETF or a bond index ETF. Model parameters could also vary greatly across the many different commodities such as Oil, Natural Gas, Gold, Silver, Corn, Wheat, etc.